Plot a Focal Taxon-Domain Trait Distribution

Description

Builds a 'ggplot2' strip + boxplot showing every raw trait value for one focal 'taxon_name x trait_domain_name' slice, with colour-coded Tukey fence flags and horizontal fence lines for both the standard (1.5x IQR) and extreme (3x IQR) thresholds.

Usage

```
plot_focal_trait_distribution(
  data_focal_trait_records,
  data_focal_trait_summary,
  focal_taxon,
  trait_domain,
  graphical_options,
  verbose = TRUE
)
```

Arguments

`data_focal_trait_records`	A tibble of raw trait observations for one focal taxon-domain slice. Must contain columns 'trait_value' (numeric), 'trait_name' (character), and 'trait_domain_name' (character).
`data_focal_trait_summary`	A single-row tibble of focal-slice QC statistics, as produced by 'write_trait_quality_control_report()'. Must contain numeric columns 'mean', 'median', 'IQR', and integer columns 'n_suspected_outliers' (integer) and 'outlier_fraction' (numeric).
`focal_taxon`	Character scalar. Name of the taxon being inspected. Used in the plot title.
`trait_domain`	Character scalar. Name of the trait domain being inspected. Used in the plot title.
`graphical_options`	Named list with elements 'width', 'height', 'units', 'dpi', and 'bg', as returned by 'load_active_config_value("graphical")'. Passed to 'ggview::canvas()'.

verbose	Logical. If 'TRUE' (default), the computed Tukey fence boundaries are printed to the console via 'cli::cli_inform()'.

Details

The function computes Q1, Q3, and IQR from 'data_focal_trait_records' and derives four fence values: - inner lower / upper: Q1 - 1.5 * IQR and Q3 + 1.5 * IQR - outer lower / upper: Q1 - 3 * IQR and Q3 + 3 * IQR

Each observation is classified as '"within fence"', '"mild outlier (1.5x IQR)"', or '"extreme outlier (3x IQR)"'. When 'data_focal_trait_records' contains more than one distinct 'trait_name', observations are faceted by 'trait_name'; otherwise the x-axis shows 'trait_domain_name'.

Plot layer order follows project conventions: 'ggplot()' -> facets -> scales -> labels -> theme -> 'ggview::canvas()' -> geoms.

Value

A 'ggplot2' object. The plot is not printed; call 'print()' on the return value to display it.

See Also

[write_trait_quality_control_report()], [load_trait_corrections()], [validate_trait_corrections()], [correct_trait_records()]

Examples

```
## Not run:

data_focal_trait_records <-

  data_traits_raw |>

  dplyr::filter(

    taxon_name == "Anacyclus clavatus",

    trait_domain_name == "Leaf Area"

  ) |>

  dplyr::arrange(trait_value)


data_focal_trait_summary <-

  data_qc_report |>

  dplyr::filter(

    taxon_name == "Anacyclus clavatus",

    trait_domain_name == "Leaf Area"

  )
```

```
graphical_options <-
  load_active_config_value("graphical")

p <-
  plot_focal_trait_distribution(
    data_focal_trait_records = data_focal_trait_records,
    data_focal_trait_summary = data_focal_trait_summary,
    focal_taxon = "Anacyclus clavatus",
    trait_domain = "Leaf Area",
    graphical_options = graphical_options
  )

base::print(p)

## End(Not run)
```